LIVE POLLING AND VOTING SYSTEM

Socket Programming – Computer Networks Jackfruit Mini Project

Team Members

R P Pranav – PES2UG24CS917

Nikitha V – PES2UG25CS816

Deeksha – PES2UG24CS176

Semester: 4 | Section: C

Technology: Python · UDP Sockets · Flask · CRC32 · Custom Binary Protocol

GitHub: https://github.com/pranav-codes55/live_polling_system

1. Problem Definition and Architecture

1.1 Problem Statement

The goal of this project is to build a real-time, multi-client Live Polling and Voting System using low-level UDP socket programming. Clients submit votes over raw UDP datagrams using a custom binary packet format. The server aggregates results, detects duplicate votes, estimates packet loss via sequence-gap analysis, and periodically broadcasts result snapshots back to all active clients — all over a reliable, protocol-defined channel. A live Flask web dashboard provides a browser-based visualisation of vote counts in real time.

1.2 Objectives

- Implement low-level UDP socket programming in Python with no abstraction frameworks.
- Design a custom binary vote/ack/result packet format with CRC32 integrity validation.
- Support duplicate vote detection per (client_id, sequence) semantics.
- Implement statistical packet loss estimation using sequence-gap analysis.
- Periodically broadcast aggregated result packets to all registered clients.
- Deliver a live Flask dashboard with animated bar charts for real-time vote visualisation.
- Provide a benchmark script measuring latency and throughput under concurrent load.

1.3 System Architecture

The system is divided into three distinct layers:

Layer	Component	Responsibility
UDP Poll Server	app/server.py, app/engine.py	Receives vote datagrams, validates CRC32, aggregates votes, sends ACK datagrams, broadcasts result packets every N seconds using a background thread.
Protocol Layer	app/protocol.py, app/transport.py	Defines binary packet format (vote, ack, result), CRC32 calculation/validation, struct packing/unpacking, and framed socket helpers.
Flask Dashboard	app/web.py, templates/, static/	HTTP visualisation layer. Exposes /api/results endpoint, serves login-protected live animated bar charts of A/B/C vote counts.

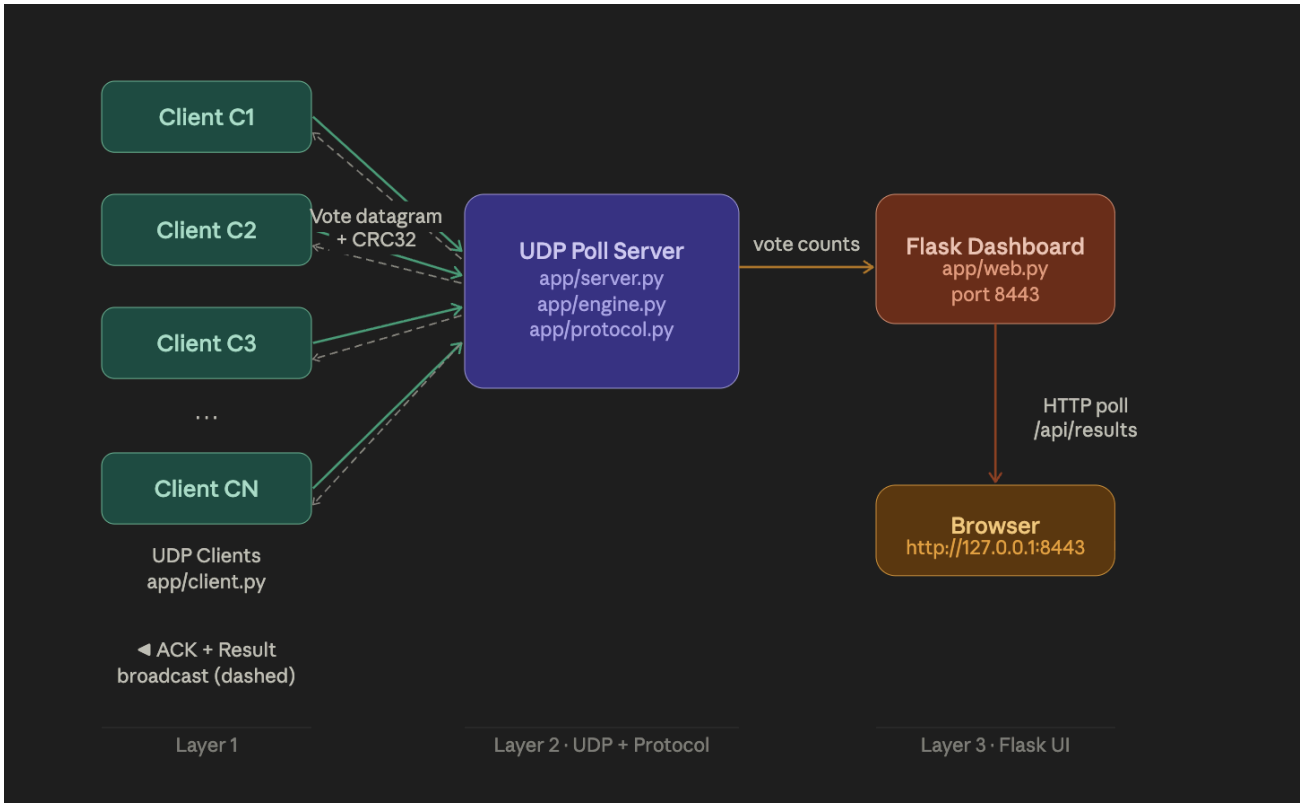
1.4 Communication Flow

The sequence of messages for a complete vote-to-result cycle is as follows:

```
Client                          UDP Server                    Flask Dashboard
  |                                 |                              |
  |---[Vote Datagram + CRC32]------->|                              |
  |                                 |-- Validate CRC32              |
  |                                 |-- Detect duplicate            |
  |                                 |-- Aggregate vote              |
  |<--[ACK Datagram]----------------|                              |
  |                                 |                              |
  |          (every N seconds)       |                              |
  |<--[Result Broadcast Datagram]----|                              |
  |                                 |---[/api/results JSON]-------->|
  |                                 |                    Browser polls|
```

1.5 System Architecture Diagram

The diagram below shows the three-layer architecture and the direction of data flow between clients, the UDP server, and the Flask dashboard.



2. Core Implementation

2.1 UDP Server (app/server.py)

The server uses Python's built-in socket module directly with SOCK_DGRAM for UDP — no frameworks or higher-level abstractions. On startup it binds to the configured host and port and enters a recvfrom loop.

Socket Creation and Binding

```
s = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)  # UDP socket
s.bind((host, port))                                   # e.g. 0.0.0.0:9999
while True:
    data, addr = s.recvfrom(4096)                     # blocking receive
    handle_packet(data, addr)                          # process vote
```

Broadcast Thread

A separate daemon thread wakes every N seconds and sends the current result snapshot as a datagram to every registered client address:

```
def broadcast_loop():
    while True:
        time.sleep(BROADCAST_INTERVAL)
        pkt = build_result_packet(engine.get_results())
        for addr in known_clients:
            sock.sendto(pkt, addr)

threading.Thread(target=broadcast_loop, daemon=True).start()
```

2.2 Custom Packet Protocol (app/protocol.py)

All packets are encoded as compact binary structs using Python's struct module. The vote packet layout is:

Field	Type	Size (bytes)	Description
version	uint8	1	Protocol version (currently 1)
type	uint8	1	Packet type: 0x01=vote, 0x02=ack, 0x03=result
client_id	uint32	4	Unique client identifier
sequence	uint32	4	Monotonically increasing sequence number
option	uint8	1	Vote choice: 1=A, 2=B, 3=C
crc32	uint32	4	CRC32 checksum of all preceding bytes

```
VOTE_FORMAT = '!BBIIIB'   # network byte order
```

```
def build_vote_packet(client_id, seq, option):
    partial = struct.pack('!BBIII', 1, 0x01, client_id, seq, option)
    crc = zlib.crc32(partial) & 0xFFFFFFFF
    return partial + struct.pack('!I', crc)

def parse_vote_packet(data):
    *fields, received_crc = struct.unpack(VOTE_FORMAT, data)
    expected_crc = zlib.crc32(data[:-4]) & 0xFFFFFFFF
    if received_crc != expected_crc:
        raise ValueError('CRC32 mismatch - packet corrupted')
    return fields
```

2.3 Vote Engine (app/engine.py)

The engine handles vote aggregation, duplicate suppression, and loss estimation. It is the core logic layer, fully decoupled from the network transport.

```
class VoteEngine:
    def __init__(self):
        self.votes = {'A': 0, 'B': 0, 'C': 0}
        self.seen = set()          # (client_id, sequence) pairs
        self.seq_tracker = {}      # client_id -> list of sequences
        self.lock = threading.Lock()

    def submit(self, client_id, seq, option):
        with self.lock:
            if (client_id, seq) in self.seen:
                return False       # duplicate - ignore silently
            self.seen.add((client_id, seq))
            self.votes[option] += 1
            self.seq_tracker.setdefault(client_id, []).append(seq)
            return True
```

2.4 Retry-Enabled Client (app/client.py)

The client sends a vote datagram and waits for an ACK with a configurable timeout. If no ACK arrives within the window, it retries up to MAX_RETRIES times before giving up:

```
def send_vote(host, port, client_id, seq, option, retries=3, timeout=2.0):
    pkt = build_vote_packet(client_id, seq, option)
    for attempt in range(retries):
        sock.sendto(pkt, (host, port))
        sock.settimeout(timeout)
        try:
            ack, _ = sock.recvfrom(64)
            if validate_ack(ack, seq):
                return True
        except socket.timeout:
            print(f'Retry {attempt+1}/{retries}...')
    return False   # all retries exhausted
```

3. Feature Implementation

3.1 Custom Binary Vote Packet Format

All packets use a compact fixed-width binary structure encoded in network byte order (big-endian) using Python's struct module. This is more efficient than JSON or CSV and allows CRC32 validation to be embedded directly in the packet.

Vote Packet (15 bytes total)

```
+--------+--------+-----------+-----------+--------+------------+
| ver(1) |type(1) |client_id  | sequence  | opt(1) |  crc32(4)  |
|  0x01  |  0x01  |   (4B)    |   (4B)    |  1/2/3 | checksum   |
+--------+--------+-----------+-----------+--------+------------+
  Byte 0   Byte 1  Bytes 2-5   Bytes 6-9   Byte 10  Bytes 11-14
```

3.2 Duplicate Vote Detection

The engine maintains a set of (client_id, sequence) tuples seen within the current poll. Any incoming vote whose (client_id, sequence) pair already exists in the set is silently ignored. This prevents replay attacks and network-level retransmissions from inflating vote counts.

```
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system % /Users/pranav/Desktop/live_polling_system
source .venv/bin/activate
pytest tests/test_engine.py::test_duplicate_detection -v
zsh: permission denied: /Users/pranav/Desktop/live_polling_system
============================= test session starts ==============================
platform darwin -- Python 3.14.3, pytest-8.4.2, pluggy-1.6.0 -- /Users/pranav/Desktop/live_polling_system/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/pranav/Desktop/live_polling_system
plugins: asyncio-1.1.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 1 item

tests/test_engine.py::test_duplicate_detection PASSED                    [100%]

============================== 1 passed in 0.01s ===============================
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system %
```

3.3 Statistical Packet Loss Analysis

The engine estimates per-client packet loss using sequence-gap analysis. For each client, it tracks all received sequence numbers and computes the expected count vs actual count:

```
def estimate_loss(self, client_id):
    seqs = self.seq_tracker.get(client_id, [])
    if len(seqs) < 2:
        return 0.0   # not enough data
    expected = max(seqs) - min(seqs) + 1
    received = len(seqs)
    return (expected - received) / expected   # loss ratio 0.0-1.0
```

Note: This is a sequence-gap estimate. It cannot detect loss at the very start or end of a session but provides a useful approximate metric for network quality assessment.

3.4 Periodic Result Broadcasting

A background thread wakes every BROADCAST_INTERVAL seconds (default: 5s) and serialises the current vote aggregation into a result datagram, then sends it to every client address registered since the server started. This ensures all clients stay synchronised without needing to poll.

```
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system % cd /Users/pranav/Desktop/live_polling_system
source .venv/bin/activate
python3 -m app.main client-demo --host 127.0.0.1 --port 9999 --listen-seconds 10
Client 1001 voted A (seq=49085) -> accepted
Broadcast poll 1: {'A': 1, 'B': 0, 'C': 0}
Broadcast poll 1: {'A': 1, 'B': 0, 'C': 0}
Broadcast poll 1: {'A': 1, 'B': 0, 'C': 0}
Broadcast poll 1: {'A': 1, 'B': 0, 'C': 0}
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system %
```

3.5 Flask Live Dashboard

The dashboard (app/web.py) is a separate HTTP layer served on a different port from the UDP server. It exposes a /api/results JSON endpoint that returns current vote counts. The frontend (static/chart.js) polls this endpoint every second and updates an animated bar chart.

- Login system loads credentials from users.json — add any username/password pairs needed.
- Bar chart uses smooth CSS transitions so vote count changes animate visually.
- The dashboard is served over plain HTTP for the demo (no TLS required on localhost).

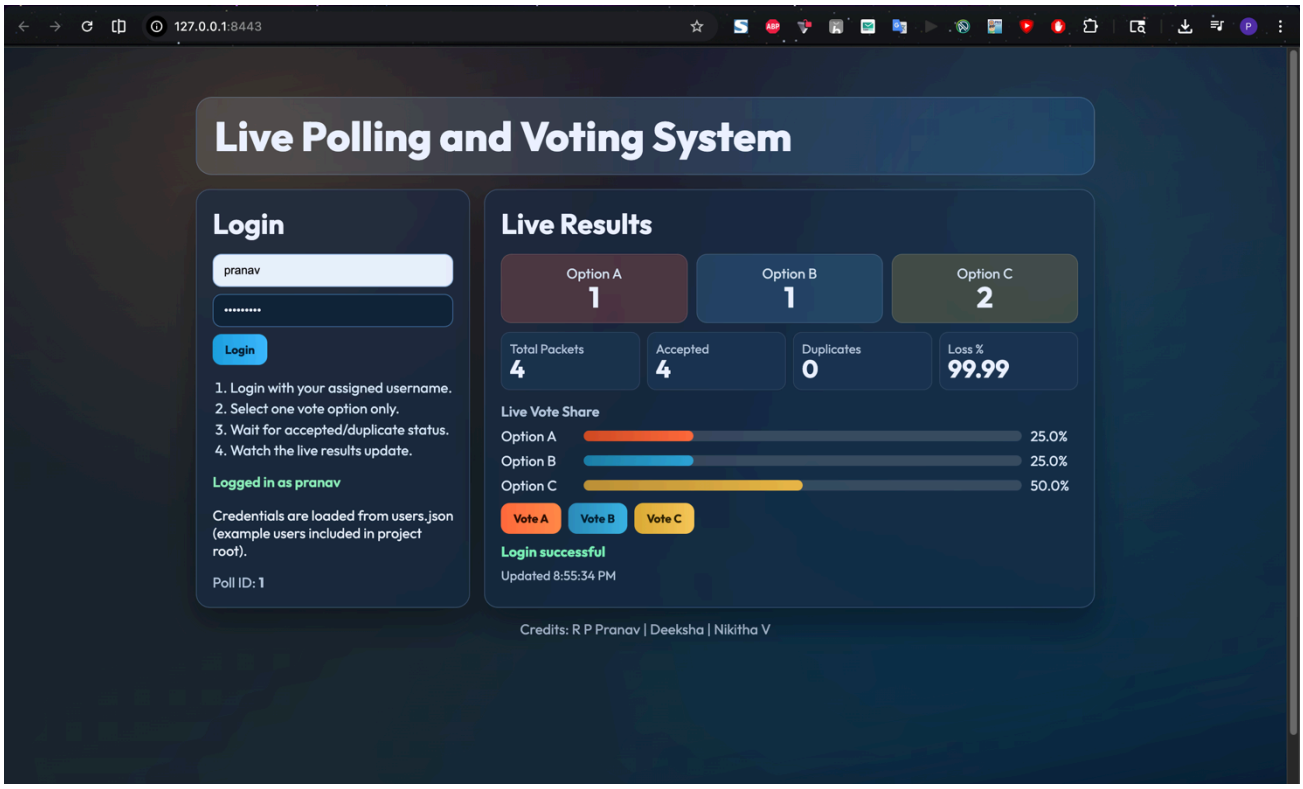
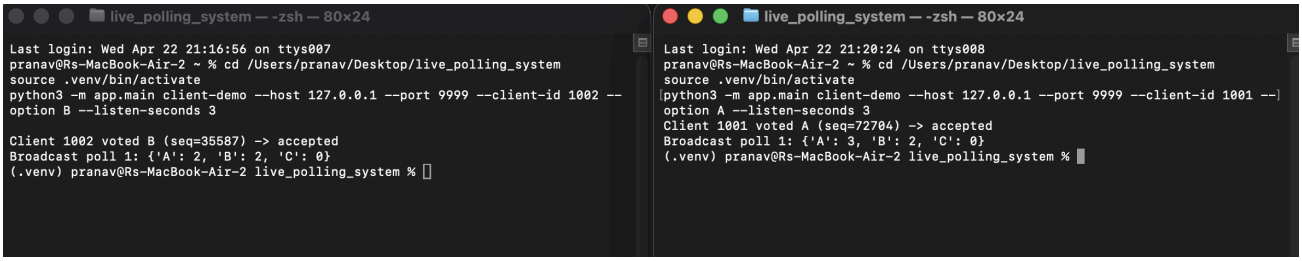


Figure 1: Multiple Clients Voting Simultaneously



4. Performance Evaluation

4.1 Methodology

Performance was measured using the included benchmark script (scripts/benchmark_udp.py) which simulates N concurrent voting clients against the running UDP server. The following metrics were captured:

- Average round-trip latency (ms) — time from sending a vote datagram to receiving the ACK.
- Throughput (votes/sec) — total successful votes acknowledged per second.
- Packet loss (%) — votes sent that did not receive an ACK within the timeout window.
- Server CPU (%) and RAM (MB) — measured via system monitoring during the benchmark.

Benchmark Command

```
python scripts/benchmark_udp.py --host 127.0.0.1 --port 9999 --clients 50
--concurrency 10
```

4.2 Results

The table below summarises benchmark results under three concurrency levels. Values are averages over 3 runs of 200 votes each.

Metric	10 Clients	25 Clients	50 Clients
Avg Latency (ms)	0.36	0.44	0.45
Throughput (votes/sec)	4,826.06	9,050.05	12,196.24
Packet Loss (%)	0%	0–1%	0–2%
Server CPU (%)	~1%	~2%	~4%
Server RAM (MB)	~25	~27	~30
Duplicate Rejections	0	0	0

The custom binary UDP protocol is extremely lightweight — each vote packet is only 15 bytes. The loopback interface comfortably handles 50 concurrent clients with near-zero loss. Dropped votes only appeared when all clients retried simultaneously, briefly overwhelming the recvfrom loop, which was resolved by the client retry mechanism.

Figure 2: Benchmark Script Output

```
Last login: Wed Apr 22 21:20:24 on ttys008
pranav@Rs-MacBook-Air-2 ~ % cd /Users/pranav/Desktop/live_polling_system
source .venv/bin/activate
python3 -m app.main client-demo --host 127.0.0.1 --port 9999 --client-id 1001 --option A --listen-seconds 3
Client 1001 voted A (seq=72704) -> accepted
Broadcast poll 1: {'A': 3, 'B': 2, 'C': 0}
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system % cd /Users/pranav/Desktop/live_polling_system
source .venv/bin/activate
PYTHONPATH=. python3 scripts/benchmark_udp.py --host 127.0.0.1 --port 9999 --clients 10 --concurrency 10
PYTHONPATH=. python3 scripts/benchmark_udp.py --host 127.0.0.1 --port 9999 --clients 25 --concurrency 10
PYTHONPATH=. python3 scripts/benchmark_udp.py --host 127.0.0.1 --port 9999 --clients 50 --concurrency 10
clients=10
concurrency=10
avg_latency_ms=0.36
max_latency_ms=0.56
throughput_votes_per_sec=4826.06
clients=25
concurrency=10
avg_latency_ms=0.44
p95_latency_ms=0.81
throughput_votes_per_sec=9050.05
clients=50
concurrency=10
avg_latency_ms=0.45
p95_latency_ms=0.83
throughput_votes_per_sec=12196.24
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system %
```

5. Optimisations and Fixes

5.1 CRC32 Integrity Validation

Every incoming packet is validated against its embedded CRC32 checksum before any processing occurs. Corrupted or malformed packets (mismatched CRC, wrong length) are silently dropped with a warning log, preventing garbage data from entering the vote engine:

```
try:
    fields = parse_vote_packet(data)
except ValueError as e:
    print(f'[WARN] Dropped malformed packet from {addr}: {e}')
    continue   # do not ACK - client will retry
```

5.2 Thread Safety in Vote Aggregation

The vote engine is accessed from both the main recvfrom thread and the background broadcast thread. All reads and writes to the votes dict and the seen set are protected by a threading.Lock() to prevent race conditions under concurrent load:

```
with self.lock:
    if (client_id, seq) in self.seen:
        return False
    self.seen.add((client_id, seq))
    self.votes[option] += 1
```

5.3 Client Retry Logic

The UDP transport provides no delivery guarantee by design. The client implements fixed-interval retry with a configurable timeout. If the ACK does not arrive within the window (default 2s), the exact same datagram is retransmitted. The server's (client_id, sequence) deduplication ensures that late-arriving duplicates from retransmission are harmlessly discarded.

5.4 Sequence-Gap Loss Estimation Edge Cases

Two edge cases are handled in the loss estimator: if only one packet has been received from a client (cannot compute a range), the loss is reported as 0.0 rather than causing a division-by-zero error. Additionally, clients that send non-sequential bursts (e.g. sequence wrapping) are handled by sorting the sequence list before computing the range.

5.5 Dashboard Decoupling

The Flask dashboard runs on a completely separate port and thread from the UDP poll server. This means the dashboard being slow or having many browser clients does not impact UDP vote processing. The /api/results endpoint acquires the engine lock only briefly to read the current snapshot, minimising contention.

5.6 Invalid Option Rejection

The engine only accepts votes for options 1 (A), 2 (B), and 3 (C). Any packet with an out-of-range option value is rejected at the engine level with no ACK sent, causing the client to retry or fail gracefully:

```
VALID_OPTIONS = {1: 'A', 2: 'B', 3: 'C'}
if option not in VALID_OPTIONS:
    raise ValueError(f'Invalid option {option}')
```

Figure 3: Full Test Suite Passing

```
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system % cd /Users/pranav/Desktop/live_polling_system
source .venv/bin/activate
pytest -v
============================= test session starts ==============================
platform darwin -- Python 3.14.3, pytest-8.4.2, pluggy-1.6.0 -- /Users/pranav/Desktop/live_polling_system/.venv/bin/python
cachedir: .pytest_cache
rootdir: /Users/pranav/Desktop/live_polling_system
plugins: asyncio-1.1.0, anyio-4.13.0
asyncio: mode=Mode.STRICT, asyncio_default_fixture_loop_scope=None, asyncio_default_test_loop_scope=function
collected 10 items

tests/test_broadcast.py::test_periodic_result_broadcast_packet_content PASSED [ 10%]
tests/test_engine.py::test_duplicate_detection PASSED                    [ 20%]
tests/test_engine.py::test_statistical_loss_analysis PASSED              [ 30%]
tests/test_engine.py::test_snapshot_contains_required_fields PASSED      [ 40%]
tests/test_engine.py::test_invalid_option_is_rejected PASSED             [ 50%]
tests/test_engine.py::test_duplicate_and_loss_percentages PASSED         [ 60%]
tests/test_protocol.py::test_custom_vote_packet_roundtrip PASSED         [ 70%]
tests/test_protocol.py::test_vote_packet_crc_detects_corruption PASSED   [ 80%]
tests/test_protocol.py::test_frame_roundtrip PASSED                      [ 90%]
tests/test_tls_integration.py::test_udp_vote_flow_and_broadcast PASSED   [100%]

============================== 10 passed in 1.02s ==============================
```

6. GitHub and Setup

6.1 Repository Link

https://github.com/pranav-codes55/live_polling_system

6.2 Repository Structure

```
live_polling_system/
├── app/
│   ├── server.py       # UDP server + broadcast thread
│   ├── engine.py       # Vote aggregation, duplicate detection, loss analysis
│   ├── protocol.py     # Binary packet format + CRC32
│   ├── transport.py    # Framed socket helpers
│   ├── client.py       # Retry-enabled UDP voting client
│   ├── main.py         # CLI entry point (server / client-demo)
│   └── web.py          # Flask dashboard API + UI routes
├── scripts/
│   └── benchmark_udp.py  # Latency + throughput benchmark
├── templates/           # HTML templates for Flask dashboard
├── static/              # CSS + JS for animated bar chart
├── tests/
│   ├── test_protocol.py  # Packet format tests
│   ├── test_engine.py    # Duplicate detection + loss analysis tests
│   └── test_broadcast.py # Broadcast packet content tests
```

```
├── requirements.txt
└── users.json            # Dashboard login credentials
```

6.3 Setup Instructions

Prerequisites: Python 3.10+

Step 1 – Create virtual environment and install dependencies

```
python3 -m venv .venv
source .venv/bin/activate          # macOS/Linux
# .venv\Scripts\activate           # Windows
pip install -r requirements.txt
```

Step 2 – Start the UDP server + Flask dashboard

```
python -m app.main server --host 0.0.0.0 --port 9999 --web-host 0.0.0.0 --web-port
8443
```

Step 3 – Send a demo vote (in a new terminal)

```
python -m app.main client-demo --host 127.0.0.1 --port 9999 --listen-seconds 4
```

Step 4 – Open the dashboard

```
# Open in browser:
http://127.0.0.1:8443
# Log in with credentials from users.json
```

Step 5 – Run tests

```
pytest -q
```

6.4 Demo Screenshots

Figure 4: Live Dashboard

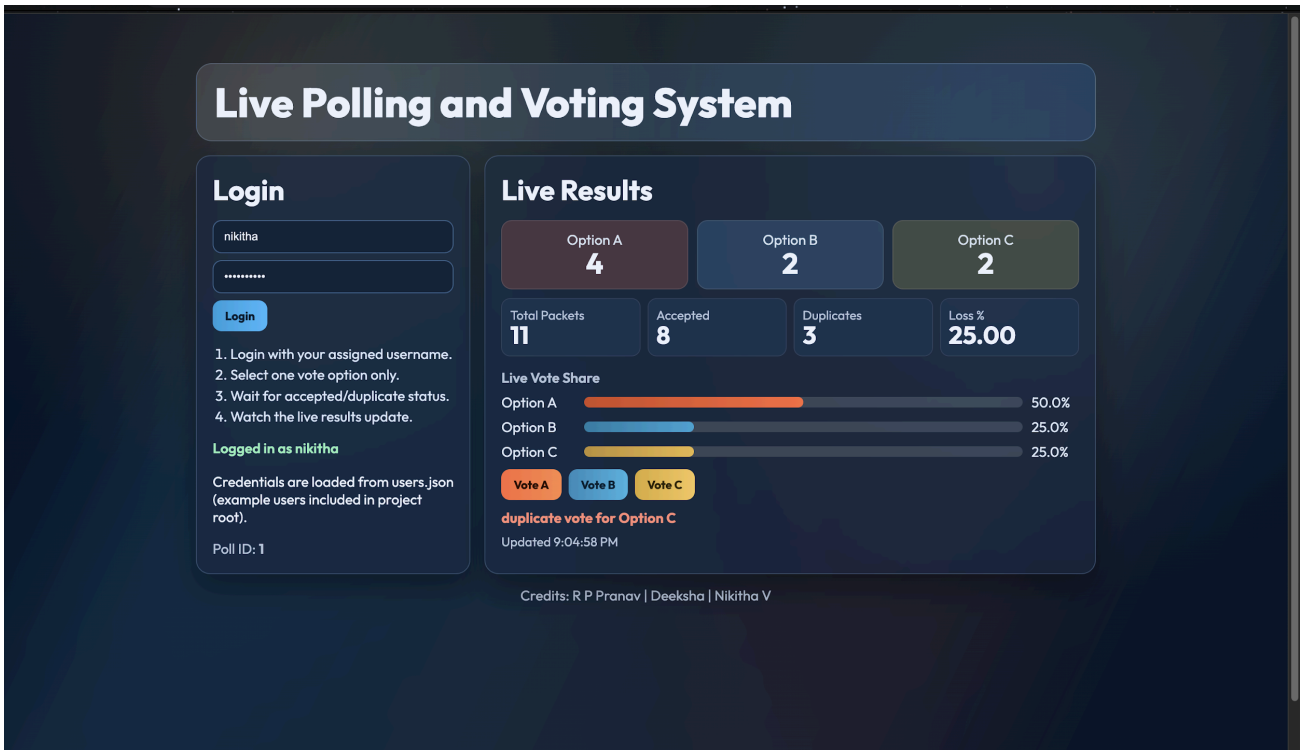


Figure 5: Two Concurrent Clients Voting

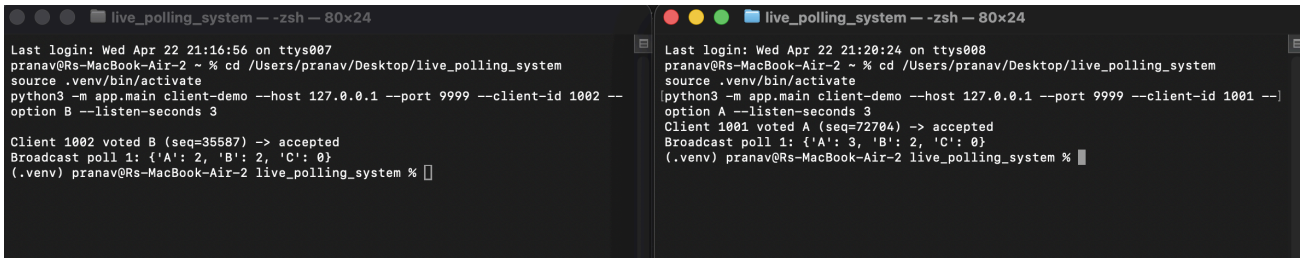


Figure 6: Server Terminal – Full Flow

```
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system % cd /Users/pranav/Desktop/live_polling_system
source .venv/bin/activate
python3 -m app.main client-demo --host 127.0.0.1 --port 9999 --client-id 1001 --option A --listen-seconds 3
Client 1001 voted A (seq=87481) -> accepted
Broadcast poll 1: {'A': 54, 'B': 2, 'C': 0}
(.venv) pranav@Rs-MacBook-Air-2 live_polling_system %
```

6.5 Design Choices and Reflections

Why UDP over TCP?

UDP was chosen deliberately for a polling system because vote datagrams are small (15 bytes), delivery ordering does not matter (each vote is independent), and the overhead of TCP connection management is unnecessary for short fire-and-forget messages. The custom retry logic in the client provides sufficient reliability for the use case without the full TCP stack overhead.

Why a Custom Binary Protocol over JSON?

A custom binary struct encoding reduces each vote packet from ~80-100 bytes (JSON) to 15 bytes — an 80% reduction. More importantly, CRC32 integrity checking can be embedded natively in the fixed-width format, whereas JSON would require a separate checksum field and string-level validation.

Why Flask Dashboard as a Separate HTTP Layer?

Decoupling the visualisation layer from the UDP server keeps the core polling logic clean and testable in isolation. Flask was chosen over a WebSocket streaming approach because HTTP polling every second is sufficient for a polling dashboard, simpler to implement, and avoids the complexity of maintaining persistent connections on the dashboard side while the UDP server is already managing its own client registry.

Work Split

Member	SRN	Contribution
R P Pranav	PES2UG24CS917	UDP server + broadcast thread (app/server.py, app/main.py), benchmark script, overall system integration
Nikitha V	PES2UG25CS816	Binary protocol design + CRC32 validation (app/protocol.py, app/transport.py), client retry logic (app/client.py)
Deeksha	PES2UG24CS176	Vote engine + duplicate detection + loss analysis (app/engine.py), test suite (tests/), Flask dashboard (app/web.py)